

flat screen display space. LED flat screens provide marvellous visual appeal, lower energy consumption, longer lifetime, improved robustness, etc," says Chaudhary.

Truly LED-powered LCD displays are the rage today. Basically, LCDs depend on some kind of backlighting and standard fluorescent bulbs were the norm earlier. Now manufacturers are slowly transitioning to LEDs for this purpose. The image quality is much better when LEDs are used. LED TVs are more environment-friendly than LCD and plasma TVs since they use much less power and no mercury in their manufacture. LEDs are also small, produce less heat and can be housed in thinner cases—the reason why you can now find really thin flat screens (5 cm). Further, LEDs are used in groups such that even if one burns out, the remaining LEDs will still produce light—this increases the life of the product.

Samsung, Sony, LG, Vizio, Panasonic, Sharp and almost all other leading brands have LED-based LCD displays.

OLED: A step ahead

Some brands like Sony and Samsung have gone a step ahead and launched organic light emitting diode (OLED) displays. OLED is a relatively new technology which is likely to replace LCDs over time—thanks to its superiority in terms of display quality, eco-friendliness and power-efficiency.

An OLED display basically comprises layers of an organic material called polymer which can emit photons and therefore light when electrons are conducted through them. There are around six variants of OLED technology as of now, and it would be exciting to watch what the brands offer with these technologies because OLED is believed to be highly suitable for high-definition TVs (HDTV). OLED-based HDTVs are sure to be high-resolution, widescreen, very thin and power-efficient. The tiny amounts of electri-



Sony's OLED TV

cal power needed to power this type of display also makes these displays 'cool'. However, OLED displays are very expensive today. So common man still has to wait to afford it.

Speed drive technology

As far as HDTV goes, everybody likes it big. However, a large size also requires speed, and hence a majority of the current generation of LCD TVs use the 100Hz speed drive technology to reach a reasonable level of dynamic definition. Basically this technology increases the image frame rate by inserting a new image between two images, through the original signals. The viewer therefore enjoys a smoother picture without any trailing even when watching fast-moving objects on TV.

As far as plasma TV goes, Panasonic raised the bar very high with its

Viera Neo PDP range which boasted of a refresh rate of 600Hz. Other players including Samsung and LG followed suit.

Internet TV: Get, set, net

Manufacturers such as LG, Panasonic, Samsung, Sharp and Vizio offer HDTVs with Ethernet connection. Models with wireless connectivity are also making an entrance. You can connect to the Internet and access Web-based content offered by the manufacturer. Common services include news and weather reports, video and movie downloads from YouTube, Netflix, etc and photo managers such as Picasa.

A recent *New York Times* report provides a good reason why you should go for an Internet-enabled TV even if you do not plan to use it for surfing: "...an Ethernet connection can prove useful when a manufacturer needs to upgrade the digital TV's operating system to, for example, address incompatibilities between it and a set-top box. Some day manufacturers will be able to use your TV's Internet connection to remotely diagnose any problems with the set." Apparently, the picture quality of Sharp TVs can be adjusted remotely over the Internet by the company's engineers.

Connecting with other devices

HDMI cables have now become the *de facto* standard for use in HDTVs as

Brand Buzz—some keywords doing the rounds

- Panasonic: Neo PDP displays which consume 50 per cent less power than other models, Internet TV, etc
- Sony: Internet connectivity and support for Yahoo! Widgets, LED backlight displays, OLED displays, etc
- LG: Thin LED backlights, wireless 1080p full HD TV, broadband-enabled TVs and energy-saving features
- Intex: Eye Safe-T Matrix monitors
- Sharp: Remote display quality management for Internet-enabled TVs
- Samsung: LED backlight displays, OLED displays, Internet TV, etc
- Toshiba (Regzas): Dolby volume, InstaPort feature to reduce lag when switching between HDMI ports and Resolution+ upscaling technology that makes standard definition content look good on a full HD screen